

Title: Development of an Artificial Intelligence (AI) chatbot to support surgeon–patient communication in thyroid cancer care

Anusha R. Talwalkar, MPH,^{a,b} Tahmid Mahbub,^c Ibrahim Ahmadli,^c Elizabeth Wendt, MD, MPH,^a Rebecca S. Sippel, MD, FACS,^a Nadine Connor, PhD,^d Alan B. McMillan, PhD,^c Courtney J. Balentine, MD, MPH, FACS^{a,b}

[a] Department of Surgery, University of Wisconsin SMPH, Madison, WI

[b] Wisconsin Surgical Outcomes Research Program, Madison, WI

[c] Department of Radiology, University of Wisconsin SMPH, Madison, WI

[d] Department of Otolaryngology, University of Wisconsin SMPH, Madison, WI

Introduction:

Thyroid cancer survivors often feel unprepared for the numerous challenges that arise during survivorship (e.g. fatigue, voice and swallow changes, long-term thyroid hormone management), which causes distress and reduces quality of life. Unfortunately, clinical time constraints limit the amount of comprehensive, on-demand counseling physicians can provide. To bridge this gap, we performed initial validation of a patient-facing large language model (LLM)-based chatbot aimed to provide supportive counseling and deliver survivorship-focused education to help patients anticipate and manage these challenges as they arise.

Methods:

To establish the initial chatbot model, commonly asked patient questions were gathered from physicians and rephrased using ChatGPT-5.2 into 177 patient-style queries with corresponding physician-level answers. These answers were manually verified for correctness by a board-certified surgeon to create a reference dataset for chatbot validation. The validated questions were programmatically passed to the chatbot, operating with a spectrum of six cost-effective LLMs (ChatGPT-4o, ChatGPT-4o-mini, ChatGPT-5-mini, ChatGPT-5-nano, Gemini-2.5-Flash, Gemini-2.5-Flash-Lite). Generated answers were assessed against the reference dataset using BLEU, ROUGE, and LLM-as-a-judge metrics to determine which model was best suited for building the chatbot.

Results:

GPT-5-mini achieved the highest overall LLM-as-a-judge score (9.05/10), followed by GPT-5-nano (8.89). Conversely, Gemini-2.5-Flash scored highest in lexical overlap metrics, leading in both BLEU and ROUGE scores. GPT-4o-mini consistently scored the lowest across all evaluations.

Conclusion:

Although Gemini-2.5-Flash achieved the highest lexical overlap, GPT-5-mini performed the best overall by delivering superior correctness, safety, and completeness. Future work will integrate Retrieval-Augmented Generation (RAG) to further enhance the chatbot's clinical performance.

Keywords: Thyroid cancer, clinical communication, artificial intelligence, patient-centered care, quality of life